ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** metal	Metal.
2	**(d)** copper	Metal.
3	**(d)** metal oxide	Oxide.
4	**(c)** non-metal	Non-metal.
5	**(d)** rubber	Non-metal.
6	**(a)** non-metal oxide	Oxide.
7	**(d)** mercury	Mercury.
8	**(b)** iron	Metal.
9	**(d)** white flame	Dazzling white.
10	**(a)** shiny	Lustre.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Iron (1); Copper (1).	1+1
12	Drawn into wires (1); Copper wire (1).	1+1
13	Sulphur (1); Rubber (1).	1+1
14	Sulphur (1); Oxygen (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lustre (1); Malleable (1); Good conductors (1).	1+1+1
16	Good conductor (1); Ductile (1); Long lasting (1).	1+1+1
17	Mercury is liquid (1); Graphite conducts (1); Sodium is soft (1).	1+1+1
	OR: Dull (1); Brittle (1); Poor conductors (1).	1+1+1
18	Malleable (1); Light (1); Non-toxic (1).	1+1+1
	OR: Bridges (1); Tools (1); Vehicles (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lustre (1); Malleability (1); Ductility (1); Conductivity (1); Sonority (1).	1+1+1+1+1
	OR: Burn sulphur (1); Gas forms (1); Dissolve in water (1); Acid forms (1); Blue litmus turns red (1).	1+1+1+1+1
20	Lustre (1); Hardness (1); Brittleness (1); Conduction (1); Sound (1).	1+1+1+1+1
	OR: Zinc and HCl (1); Magnesium and HCl (1); Hydrogen released (1); Salt forms (1); Pop test (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that all metals are hard and solid; **mercury** is liquid and **sodium** is soft.
- Saying that all non-metals are poor conductors; **graphite** conducts electricity.
- Mixing up oxides: metal oxides are *basic* (red litmus to blue), non-metal oxides are *acidic* (blue litmus to red).
- Confusing malleability (sheets) with ductility (wires).
- Forgetting that rusting needs **both** air and moisture, and that galvanisation uses *zinc*.
- Writing that a less reactive metal can displace a more reactive one from its solution (e.g. copper displacing iron).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** sulphur	Brittle.
2	**(c)** blue flame	Blue.
3	**(d)** dull	Dull.
4	**(b)** aluminium	Malleable.
5	**(c)** basic	Basic.
6	**(b)** copper	Copper.
7	**(a)** copper	Ductile.
8	**(b)** acidic	Acidic.
9	**(d)** aluminium	Aluminium.
10	**(b)** silver	Silver.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ringing sound (1); Bells (1).	1+1
12	Basic (1); Turns red litmus blue (1).	1+1
13	Shine of metals (1); Seen in gold (1).	1+1
14	Mercury (1); At room temperature (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Utensils (1); Wires (1); Machines (1).	1+1+1
16	White flame (1); White powder (1); Magnesium oxide (1).	1+1+1
17	Conduct heat (1); Burn hands (1); Use plastic or wood (1).	1+1+1
	OR: Oxygen to breathe (1); Carbon as fuel (1); Sulphur in matches (1).	1+1+1
18	Blue flame (1); Sharp smelling gas (1); Sulphur dioxide (1).	1+1+1
	OR: Look for shine (1); Test conduction (1); Hammer it (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Wires (1); Utensils (1); Bridges (1); Jewellery (1); Machines (1).	1+1+1+1+1
	OR: Mercury (1); Sodium (1); Graphite (1); Iodine (1); Diamond (1).	1+1+1+1+1
20	Oxygen: breathing (1); Carbon: fuel (1); Sulphur: matches (1); Nitrogen: fertilisers (1); Chlorine: water treatment (1).	1+1+1+1+1
	OR: Conduct heat (1); Strong (1); Can be shaped (1); Do not break (1); Easy to clean (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that all metals are hard and solid; **mercury** is liquid and **sodium** is soft.
- Saying that all non-metals are poor conductors; **graphite** conducts electricity.
- Mixing up oxides: metal oxides are *basic* (red litmus to blue), non-metal oxides are *acidic* (blue litmus to red).
- Confusing malleability (sheets) with ductility (wires).
- Forgetting that rusting needs **both** air and moisture, and that galvanisation uses *zinc*.
- Writing that a less reactive metal can displace a more reactive one from its solution (e.g. copper displacing iron).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** hydrogen	Hydrogen.
2	**(d)** gold	Gold.
3	**(d)** graphite	Carbon.
4	**(b)** pop sound	Pop.
5	**(d)** metals	Sonorous.
6	**(d)** hardest substance	Carbon form.
7	**(c)** brown	Copper.
8	**(d)** oxygen	Non-metal.
9	**(d)** metals	Conduct heat.
10	**(a)** kerosene	Reactive.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Acidic (1); Turns blue litmus red (1).	1+1
12	Beaten into sheets (1); Aluminium foil (1).	1+1
13	Copper (1); Silver (1).	1+1
14	Hydrogen (1); Burns with a pop (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Gold (1); Silver (1); Iron (1).	1+1+1
16	$Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Hydrogen evolved (1); Zinc chloride formed (1).	1+1+1
17	Wires (1); Vessels (1); Coins (1).	1+1+1
	OR: Carbon (1); Sulphur (1); Oxygen (1).	1+1+1
18	Gold (1); Silver (1); Aluminium (1).	1+1+1
	OR: Foil (1); Utensils (1); Window frames (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Make a circuit (1); Leave a gap (1); Place the material (1); Bulb glows for a conductor (1); No glow for an insulator (1).	1+1+1+1+1
	OR: Use a magnet for iron (1); Check lustre (1); Check conduction (1); Hammer test (1); Sound (1).	1+1+1+1+1
20	Heat the metal (1); It burns (1); Oxide forms (1); Magnesium makes white powder (1); Oxide is basic (1).	1+1+1+1+1
	OR: Dip nail (1); Brown copper (1); Blue fades (1); Pale green solution (1); Iron is more reactive (1).	1+1+1+1+1

Mistakes to avoid:

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- Saying that all non-metals are poor conductors; **graphite** conducts electricity.
- Mixing up oxides: metal oxides are *basic* (red litmus to blue), non-metal oxides are *acidic* (blue litmus to red).
- Confusing malleability (sheets) with ductility (wires).
- Forgetting that rusting needs **both** air and moisture, and that galvanisation uses *zinc*.
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